



University of Colombo, Sri Lanka

University of Colombo School of Computing

BACHELOR OF SCIENCE IN INFORMATION SYSTEMS

First Year Examination - Semester I - UCSC AY22[held in Sept./ Oct. 2025]

IS1206 — Mathematics for Computing

(Two (2) Hours)

Answer ALL questions

Number of Pages = 15

Number of Questions = 4



To be completed by the candidate

Index Number:

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Important Instructions to Candidates:

- I. Students should answer in the medium of English language only using the space provided in this question paper.
- II. Note that questions appear on both sides of the paper. If a page or a part of this question paper is not printed, please inform the supervisor immediately.
- III. Write your index number CLEARLY on each and every page of this Question paper.
- IV. This paper consists of 4 questions in 15 pages (including the Cover Page).
- V. Answer ALL questions.
- VI. Calculators and any electronic device capable of storing and retrieving text, including electronic dictionaries, smartwatches, and mobile phones, are not allowed.
- VII. Do not tear off any part of this answer book. Under no circumstances may a candidate remove this book, used or unused, from the Examination Hall.

To be completed by the examiners

1	
2	
3	
4	
Total	

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1.

- [3 Marks]

[illegible]

- $$\log_a x = \frac{1}{2} [\log_a 9 + \log_a 12 - \log_a 3]$$

[5 Marks]

1. The first part of the document is a title page. It contains the title of the report, the author's name, and the date of the report. The title is "The Effect of Temperature on the Rate of Reaction of Hydrogen Peroxide with Potassium Iodide". The author is "John Doe". The date is "10/10/2023".

2. The second part of the document is an abstract. It provides a brief summary of the purpose of the experiment, the methods used, and the results obtained. The abstract states that the purpose of the experiment was to determine the effect of temperature on the rate of reaction of hydrogen peroxide with potassium iodide. The methods used were a series of experiments at different temperatures, and the results showed that the rate of reaction increased with increasing temperature.

3. The third part of the document is the introduction. It provides background information on the reaction being studied, and explains why it is important to study the effect of temperature on the rate of reaction. The introduction states that the reaction between hydrogen peroxide and potassium iodide is a classic example of a reaction that is affected by temperature. It is important to study this reaction because it helps to understand the general principles of chemical kinetics.

4. The fourth part of the document is the experimental procedure. It describes the steps that were followed during the experiment, including the preparation of the reactants, the measurement of the rate of reaction, and the control of the temperature. The procedure states that a series of experiments were conducted at different temperatures, and the rate of reaction was measured by the time taken for a certain amount of iodine to be produced.

5. The fifth part of the document is the results and discussion. It presents the data obtained from the experiments, and discusses the results in terms of the effect of temperature on the rate of reaction. The results show that the rate of reaction increased with increasing temperature, and the discussion explains this in terms of the activation energy of the reaction.

6. The sixth part of the document is the conclusion. It summarizes the findings of the experiment, and states the overall conclusion. The conclusion states that the rate of reaction of hydrogen peroxide with potassium iodide increases with increasing temperature.

7. The seventh part of the document is the references. It lists the sources of information used in the report, including textbooks, journal articles, and websites. The references are listed in alphabetical order.

8. The eighth part of the document is the appendix. It contains any additional information that is relevant to the report, such as raw data, calculations, and diagrams. The appendix is organized into sections, and each section is labeled with a number and a title.

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(c). Simplify the following arithmetic expression:

$$\frac{(45 + 72 \div 6 \times 2 - 18)}{\sqrt{25}} \times (|-8 + 3|) - 20 \div \left(\frac{8}{2}\right) - 2 \times 13$$

[8 Marks]

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(d). In a computer lab, three periodic maintenance tasks run at different intervals:

- Task A: every 84 minutes
- Task B: every 126 minutes
- Task C: every 210 minutes

(i). Find the **Highest Common Factor (HCF)** of the above three task intervals.

[4 Marks]

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(ii). Explain the practical significance of the HCF in scheduling the maintenance tasks above.

[2 Marks]

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(iii). Find the **Least Common Multiple (LCM)** of the three task intervals above.

[3 Marks]

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2.

(a). State whether each of the following statements is **TRUE** or **FALSE**.

If a statement is **FALSE**, write the correct equivalent statement in the space provided.

If a statement is **TRUE**, provide a brief justification for your answer.

Note: No marks will be provided if a justification for a TRUE statement or the correct equivalent statement for a FALSE statement is not provided..

[8 Marks]

- (i). Roster form representation of the statement '*The set of all positive integers that are perfect squares*' is $X = \{x: x \in \mathbb{Z}^+, \exists n \in \mathbb{Z}^+ \text{ such that } x = n^2\}$

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- (ii). Suppose $A = \{x \in \mathbb{Z} \mid -3 \leq x \leq 3\}$; $B = \{-3, -2, -1, 0, 1, 2, 3\}$;

$$C = \{x \in \mathbb{N} \mid 1 \leq x \leq 7\}.$$

The Sets A and B are **equal**, and A and C are **equivalent**.

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- (iii). If $A \subseteq B$, then every element in B must also be in A.

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- (iv). The symmetric difference of two sets A and B contains all elements that are in both A and B.

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- (b). Let A, B, and C be sets. Show that the following identity holds by applying the **laws of set theory**.

$$((A \cap B^c) \cup (B - C))^c = (A^c \cap B^c) \cup (A^c \cap C) \cup (B \cap C)$$

[6 Marks]

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- (c). A company is testing a new software system. Suppose the following propositions are defined:

p : "The software passes all unit tests."

q : "The software is deployed to production."

r : "Users report system errors."

- (i). Express the statement **"If the software passes all unit tests and is deployed to production, then users will not report system errors"** using p , q , and r logical connectives.

[3 Marks]

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- [8 Marks]

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3.

(a). A food delivery application assigns delivery fees based on the distance from the restaurant to the customer, using the following rules.

For distances $d \leq 3\text{ km}$, the delivery fee is Rs. 150.

For $3 < d \leq 8\text{ km}$, the delivery fee is Rs. 300.

For $d > 8\text{ km}$, the delivery fee is Rs. 500.

(i). Define the delivery fee as a function of d , $f(d)$, using proper **notations**.

[4 Marks]

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(ii). Find $f(2)$, $f(5)$, and $f(12)$.

[3 Marks]

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(iii). Determine whether the function $f(d)$ is **One-to-One**. Justify your answer

[2 Mark]

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- (iv). Suppose the app calculates the **delivery distance** based on **time taken** t (in minutes) using an approximate linear relation as defined below.

$$g(t) = 0.5t$$

Define the composition $f \circ g$.

[2 Marks]

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- (b). Suppose a software company is assigning **15 identical external hard drives** to **5 project teams** (Team A, Team B, Team C, Team D and Team E).

- (i). In how many ways can the hard drives be distributed if each team can receive **any number of hard drives** (including zero)?

[3 Marks]

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- (ii). If each team must receive **at least 2 hard drives**, in how many ways can the distribution be made?

[3 Marks]

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Index Number:

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(c). Suppose a multinational company connects **three branch offices: Office P, Office Q, and Office R**. All communication from **Office P** to **Office R** must pass through **Office Q**, but some messages can optionally be routed through **Office S** for additional encryption.

- Between **P and Q**, there are **5 independent secure channels**.
- Between **Q and R**, there are **4 independent high-speed fiber links**.
- Between **Q and S**, there are **2 independent backup channels**, and between **S and R**, there are **3 independent backup channels**.

(i). How many distinct complete communication paths exist from P to R if messages do not go through S?

[2 Marks]

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(ii). How many distinct complete paths exist from P to R if messages may optionally pass through S (i.e., either via $Q \rightarrow R$ or $Q \rightarrow S \rightarrow R$)?

[4 Marks]

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- [2 Marks]

(a). A software company tracks the number of **tasks completed by three developers** over four weeks. Let the data be represented as a **matrix A**, where each row represents a developer (Row 1 $\rightarrow$ Developer 1; Row 2 $\rightarrow$ Developer 2; Row 3 $\rightarrow$ Developer) and each column represents a week:

$$A = \begin{bmatrix} 5 & 3 & 5 & 5 \\ 4 & 2 & 6 & 6 \\ 6 & 1 & 7 & 2 \end{bmatrix}$$

Let v be a vector representing the weightage of each week's tasks for calculating the performance score:

$$v = \begin{bmatrix} 1 \\ 2 \\ 2 \\ 3 \end{bmatrix}$$

- [4 Marks]

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- [4 Marks]

(b). Suppose $A = \begin{pmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{pmatrix}$ and $B = \frac{1}{4} \begin{pmatrix} 3 & -1 & -1 \\ -1 & 3 & -1 \\ -1 & -1 & 3 \end{pmatrix}$

- [3 Marks]

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11-11-11

[2 Marks]

[illegible]

[3 Marks]

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$$3x + y + z = 7$$

[2 Marks]

[7 Marks]

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